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Hierarchical Tactical Corridor Planning and Control for Wheel-to-Wheel Autonomous Racing

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This work was developed in the context of the Abu Dhabi Autonomous Racing League (A2RL) on the Yas Marina Circuit using the Dallara EAV24 platform.

Keywords: Autonomous racing, model predictive contouring control, second-order cone programming, reinforcement learning, Frenet frame, friction circle, game-theoretic motion planning.

Abstract
Wheel-to-wheel autonomous racing couples strategic interaction, real-time trajectory optimization, and friction-limit control. Solving these decisions in a single monolithic game or nonlinear program is computationally fragile at Formula-class speeds, whereas end-to-end neural control is difficult to audit on a safety-critical vehicle. This paper presents a hierarchical architecture that restricts learning to tactical corridor synthesis and retains optimization-based planning and control downstream. A map-conditioned Truncated Quantile Critics policy and a hysteretic finite-state machine select an overtake side, velocity cap, and time-varying Frenet corridor. A sparse second-order cone program computes a curvature-regularized lateral path within that corridor, a forward–backward sweep imposes a super-elliptic GG friction envelope, and a nonlinear model predictive contouring controller tracks the resulting reference with a terminal progress reward and a soft rear-axle friction constraint. Closed-loop simulations on the Dallara EAV24 at Yas Marina demonstrate two successful fast-closure overtakes, with pass times of 1.13 s and 8.63 s after entering OVERTAKE, peak speed up to 73.10 m/s, and lateral RMS tracking errors of 0.02 and 0.05 m. The results support the central premise that learning should shape the feasible set, while model-based layers enforce vehicle dynamics and handling-limit execution.

1 Introduction

Autonomous racing at the vehicle handling limit is among the most demanding closed-loop control problems in robotics. A Formula-class car must decide *whether* to overtake, *which side* to attempt, *where* to place the resulting trajectory within the track boundaries, and *how* to execute it at the friction limit—all while an adversarial opponent simultaneously optimizes its own plan. These four decisions span discrete game theory, continuous trajectory optimization, and high-bandwidth nonlinear control, operating across timescales that differ by three orders of magnitude. The Abu Dhabi Autonomous Racing League (A2RL), with its 5.28 km Yas Marina Formula 1 circuit and the 700 kg Dallara EAV24 platform, provides a uniquely demanding testbed for this hierarchical challenge [9–11].

Classical autonomous driving architectures decompose the problem into perception, behavior planning, motion planning, and control [37, 38]. The behavior planner selects a discrete maneuver—follow, change lane, yield—which the motion planner translates into a continuous trajectory [42, 43]. In racing, however, the motion planner has traditionally been cast as a single-vehicle, offline lap-time optimization problem [1–5], producing a fixed raceline tracked by an MPC or MPCC controller [6, 7, 13, 15, 16]. Full-stack racing platforms developed for the Indy Autonomous Challenge and A2RL document this paradigm in detail [8, 10–12]. The fundamental limitation is that a pre-computed raceline cannot adapt to an opponent who is simultaneously optimizing their own trajectory: an adversary that defends the apex creates a geometric deadlock that a single-vehicle planner cannot resolve without a higher-level instruction about which side to attempt the pass and when to commit.

Recognising that wheel-to-wheel racing is inherently a non-cooperative dynamic game, several groups have proposed game-theoretic formulations of the tactical layer. Wang *et al.* [21] cast head-to-head racing as an iterative best-response game with local Nash convergence guarantees. Iterative linear-quadratic game solvers [23, 24] compute approximate generalized feedback Nash equilibria at near-real-time rates. Real-time game-theoretic planners [25] and viability-theoretic approaches [16, 26] provide formal safety certificates, while sampling-based MPPI [27] and Bayesian intent inference [29, 36] address opponent-model uncertainty. Despite this progress, pure game-theoretic methods face four practical obstacles in Formula-class deployment: (i) exact Nash equilibria of continuous-time differential games are intractable, and approximate solvers [23, 24] may not converge within a single planning cycle at circuit speeds; (ii) the Nash solution is fragile to misspecification of the opponent's cost function; (iii) the discrete decision to switch from shadowing to overtaking is not naturally captured by smooth equilibrium computations; and (iv) hand-tuned weights and horizon lengths must be recalibrated for each track and pace configuration [39, 40].

In parallel, data-driven methods have emerged as an alternative. Sony's GT Sophy [28] and subsequent RL-for-racing work [30, 31, 41] demonstrated that end-to-end neural policies can achieve superhuman performance in simulation. Model-based RL [22, 33], safe RL [32, 34, 45], and learning-from-demonstration [44, 47] have further expanded the data-driven toolkit. Lin *et al.* [46] compared optimal control and RL at the handling limit, finding that RL excels in multi-agent settings but underperforms in single-vehicle lap time. However, deploying end-to-end neural controllers on a physical Formula car raises four serious concerns: (i) the EAV24's nonlinear tire model, aerodynamic downforce map, and differential torque vectoring are not faithfully reproduced in open simulators; (ii) a neural network that outputs steering torque cannot be audited or certified by race engineers; (iii) actor-critic policies produce high-frequency command jitter that degrades tire wear; and (iv) without an explicit spatial representation, the network must re-discover track geometry from reward signals alone [13, 47].

We resolve these tensions through architectural separation: the RL policy is restricted to the tactical layer, leaving model-based solvers for planning and control. A Truncated Quantile Critics (TQC) policy outputs a low-dimensional action vector that reshapes the feasible lateral corridor $\mathcal{C}_k = [L_{\min,k}, L_{\max,k}]$ and the FSM hysteresis thresholds, without ever modifying the planner cost or the controller dynamics. Critically, the TQC observation vector embeds a *fixed-track prior*: a frozen, map-level encoding of raceline curvature, track banking, and friction-limit speed at each arc-length station. This spatial prior supplies the policy with geometric context, so the learned tactical decision is conditioned on upcoming curvature and available track width rather than on reward history alone. A heuristic FSM translates the continuous TQC output into a discrete mode sequence and a corridor-carving decision, delivering a constraint triple $(\mathcal{C}_k, v_{\mathrm{cap}}, \texttt{mode})$ to the planner.

Given the corridor, the planning layer solves a sparse second-order cone program (SOCP) for a curvature-smooth lateral path, followed by a forward–backward speed sweep on a super-elliptic GG diagram with empirically identified exponent $p = 0.75$. This two-stage decomposition is substantially faster than monolithic nonlinear trajectory optimization [5] while naturally capturing the coupled longitudinal–lateral grip limits. The resulting reference is tracked by a nonlinear Model Predictive Contouring Controller (MPCC) that augments an 8-state dynamic bicycle model with a path-progress state; a terminal progress reward automatically generates corner entry-braking and exit-acceleration without per-corner reference shaping, and a soft rear-axle friction-cone constraint prevents tire saturation without rendering the nonlinear program infeasible. All three layers share a common Frenet representation.

The contributions of this paper are:

1. **Map-conditioned tactical corridor synthesis.** A TQC policy receives a frozen preview of curvature, track width, and friction-limited speed and outputs only interpretable tactical quantities: side preference, safety margin, velocity cap, and corridor bounds. A hysteretic FSM converts this output into a time-varying feasible set $\mathcal{C}_k$, preventing the learned policy from directly commanding actuators or altering downstream costs.

2. **Convex local path planning with GG-constrained speed profiling.** A sparse SOCP over Frenet state and curvature slack produces a curvature-regularized path inside the tactical corridor. A forward–backward sweep on a super-elliptic GG envelope then constructs the corresponding grip-limited speed profile, separating geometric feasibility from speed feasibility while preserving their coupling through curvature.

3. **Closed-loop validation of the hierarchy in head-to-head racing.** An 8-state dynamic-bicycle MPCC with a terminal progress reward and a soft rear-axle friction constraint executes the planned references. Two closed-loop Yas Marina fast-closure overtake simulations quantify pass time, tactical gain, peak speed, and tracking error, showing that the same planner and controller can execute distinct tactical corridors without case-specific retuning.

The remainder of the paper is organized as follows. Section 2 derives the unified vehicle and Frenet models. Section 3 presents the RL-driven tactical layer. Section 4 details the SOCP path planner and the FB speed profiler. Section 5 formulates the MPCC controller. Section 6 discusses inter-layer signaling. Section 7 reports two closed-loop overtake simulations. Section 8 concludes.

2 Vehicle and Frenet Modeling

We adopt the convention that the controller sees a body-frame dynamic single-track model (used inside MPCC), while the planner sees a Frenet kinematic model derived from the same physical vehicle. Both reduce to a common parameterization in the limit of zero side-slip.

2.1 Body-Frame Dynamic Single-Track Model

Let $\boldsymbol{x} \in \mathbb{R}^8$ and $\boldsymbol{u} \in \mathbb{R}^2$ denote the MPCC state and input,

$$\boldsymbol{x} = \left[p_x,\ p_y,\ \psi,\ v_x,\ v_y,\ r,\ \delta,\ s\right]^\top, \qquad \boldsymbol{u} = \left[a,\ \dot{\delta}\right]^\top, \tag{1}$$

where (p_x, p_y) is the position in the inertial frame attached to the *prediction window origin*, ψ is the yaw angle, (v_x, v_y) are the body-frame longitudinal and lateral velocities, $r = \dot{\psi}$ is the yaw rate, δ is the front steering angle, s is an integrated longitudinal-progress state used by the contouring cost, a is the commanded longitudinal acceleration and $\dot{\delta}$ is the steering rate. The progress state is integrated as $\dot{s} = v_x$, so s acts as the path parameter inside the contouring formulation of Sec. 5.

The kinematic relations are

$$\dot{p}_x = v_x \cos\psi - v_y \sin\psi, \quad \dot{p}_y = v_x \sin\psi + v_y \cos\psi, \quad \dot{\psi} = r. \tag{2}$$

Front and rear slip angles are obtained with the standard small-angle single-track expressions, regularized at low speed by an offset $\varepsilon_v = 0.5\,\mathrm{m/s}$,

$$\alpha_f = \arctan\left(\frac{v_y + l_f r}{v_x + \varepsilon_v}\right) - \delta, \tag{3}$$

$$\alpha_r = \arctan\left(\frac{v_y - l_r r}{v_x + \varepsilon_v}\right). \tag{4}$$

Linear cornering stiffness yields the lateral tire forces

$$F_{yf} = -C_f\,\alpha_f, \qquad F_{yr} = -C_r\,\alpha_r. \tag{5}$$

Substituting into Newton–Euler gives the dynamic equations

$$\dot{v}_x = a + r\,v_y - \tfrac{1}{m} F_{yf} \sin\delta, \tag{6}$$

$$\dot{v}_y = \tfrac{1}{m}\big(F_{yf}\cos\delta + F_{yr}\big) - r\,v_x, \tag{7}$$

$$\dot{r} = \tfrac{1}{I_z}\big(l_f F_{yf}\cos\delta - l_r F_{yr}\big), \tag{8}$$

$$\dot{\delta} = \dot{\delta}. \tag{9}$$

The full continuous dynamics are written compactly as

$$\dot{\boldsymbol{x}} = f_c(\boldsymbol{x}, \boldsymbol{u}), \tag{10}$$

and discretized with explicit Runge–Kutta of order four with sample $T_s = 0.05\,\mathrm{s}$,

$$\boldsymbol{x}_{k+1} = f_d(\boldsymbol{x}_k, \boldsymbol{u}_k) \ \triangleq\ \mathrm{RK4}\big(f_c, \boldsymbol{x}_k, \boldsymbol{u}_k, T_s\big). \tag{11}$$

Vehicle parameters are summarized in Table 1.

2.2 Frenet Kinematic Model and GG Envelope

The reference centerline (or raceline) is a smooth curve $\boldsymbol{r}(s) \in \mathbb{R}^2$ parameterized by arc-length $s \in [0, L_{\mathrm{trk}})$, with curvature $\kappa(s)$ and tangent angle $\theta(s)$ such that $\theta'(s) = \kappa(s)$. Any point in a tube around $\boldsymbol{r}$ admits the Frenet decomposition

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \boldsymbol{r}(s) + l\,\boldsymbol{n}(s), \qquad \boldsymbol{n}(s) = \big[-\sin\theta(s),\ \cos\theta(s)\big]^\top, \tag{12}$$

Table 1: Dallara EAV24 model parameters used throughout this paper.

Symbol	Value	Description
m	742 kg	Total mass
l_f, l_r	1.75, 1.35 m	Axle distances to CoG
I_z	1500 kg m^2	Yaw inertia
C_f, C_r	2.30, 2.40 $\times 10^5$ N/rad	Tire stiffness
μ	1.20	Tire–road friction
$\delta_{\max}$	0.342 rad	Steering bound ($\pm 19.6°$)
$\dot{\delta}_{\max}$	0.5 rad/s	Steering rate bound
$a \in [\underline{a}, \overline{a}]$	$[-5, 8]$ m/s^2	Long. accel. range
$v_{x,\max}$	90 m/s	Long. speed bound
$c_{d,2}$	-1.04765 N s^2/m^2	Drag coeff. (v^2)
$c_{d,1}$	47.5 N s/m	Drag coeff. (v)
$c_{d,0}$	-1609.73 N	Drag coeff. (constant)

where l is the signed lateral offset and $\alpha \triangleq \psi - \theta(s)$ the heading deviation. Differentiating (12) and projecting onto the tangent/normal frame yields, for a nonholonomic body with body speed v,

$$\dot{s} = \frac{v \cos \alpha}{1 - l\,\kappa(s)}, \tag{13}$$

$$\dot{l} = v \sin \alpha, \tag{14}$$

$$\dot{\alpha} = \dot{\psi} - \kappa(s)\,\dot{s}. \tag{15}$$

Spatial reformulation Replacing time by arc-length and using $\frac{d}{ds} = \frac{1}{\dot{s}}\frac{d}{dt}$,

$$l'(s) = \frac{(1 - l\kappa)\tan \alpha}{1}, \qquad \alpha'(s) = \frac{(1 - l\kappa)\,\dot{\psi}}{v \cos \alpha} - \kappa(s). \tag{16}$$

Linearizing (16) around a nominal $(l_i, \alpha_i, \kappa_i)$ provides the planner-side dynamics

$$\frac{d}{ds}\begin{bmatrix} l \\ \alpha \end{bmatrix} \approx \underbrace{\begin{bmatrix} -\kappa_i \tan \alpha_i & \frac{1-\kappa_i l_i}{\cos^2 \alpha_i} \\ 0 & 0 \end{bmatrix}}_{\boldsymbol{A}_i} \begin{bmatrix} l \\ \alpha \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ \frac{1-\kappa_i l_i}{v_{\text{ref}}^2 \cos \alpha_i} & 0 \end{bmatrix}}_{\boldsymbol{B}_i} \begin{bmatrix} \zeta \\ \sigma \end{bmatrix} + \boldsymbol{b}_i, \tag{17}$$

which is the building block of the SOCP discretization in Sec. 4. The auxiliary variables (ζ, σ) are the curvature slack and its second-order cone epigraph variable.

GG envelope The combined-slip envelope of the EAV24 is well captured by the super-elliptic relation

$$\left|\frac{a_y}{a_y^{\max}(v)}\right|^p + \left|\frac{a_x}{a_x^{\max}(v)}\right|^p \leq 1, \tag{18}$$

where $a_y^{\max}(v)$ and $a_x^{\max}(v)$ are speed-dependent grip caps obtained from a vehicle data table, and $p = 0.75$ is identified empirically from on-track lateral/longitudinal traces. Solving (18) for the available longitudinal acceleration gives

$$a_x^{\text{avail}}(v, a_y) = a_x^{\max}(v)\left[1 - \left(\tfrac{|a_y|}{a_y^{\max}(v)}\right)^p\right]^{1/p}, \tag{19}$$

which is the central nonlinearity of the FB speed planner. Aerodynamic drag is modeled as a quadratic function of speed,

$$a_{\text{drag}}(v) = \tfrac{1}{m}\left(c_{d,2}\,v^2 + c_{d,1}\,v + c_{d,0}\right), \tag{20}$$

whose coefficients $(c_{d,2}, c_{d,1}, c_{d,0})$ are identified from coastdown data and listed in Table 1. Drag is added to the longitudinal force balance with the sign convention of (6).

Table 2: Characteristic ranges used to normalize the tactical observation vector.

Symbol	Value
V_{char}	80 m/s
n_{char}	8 m
a_{char}	20 m/s^2
Δs_{char}	50 m
$\Delta V_{s,\text{char}}$	30 m/s
g_{char}	50 m
$\tau_{\text{tc,max}}$	30 s
ε_τ	0.01 m/s

3 Tactical Layer: RL-Driven Corridor Synthesis

The tactical layer transforms the multi-agent racing scene into a spatio-temporally smooth feasible corridor for the planner. It is the only layer that observes opponents, and it communicates with the optimizer through constraints rather than through cost weights or low-level commands. This interface keeps the learned component interpretable: changing the policy changes where the optimizer is allowed to drive, not how the optimizer values tracking, smoothness, or vehicle dynamics.

3.1 Hybrid TQC Policy, Track Prior and Finite-State Machine

Observation At each tactical step the environment supplies a compact dynamic observation $\boldsymbol{o}_t \in \mathbb{R}^{27}$. The ego-vehicle block comprises normalized speed, heading error, lateral offset, and body-frame accelerations,

$$\boldsymbol{o}_t^{\text{ego}} = \left[\bar{V},\ \chi,\ \bar{n},\ \bar{a}_x,\ \bar{a}_y,\ \bar{s}_e\right]^\top, \tag{21}$$

where each component is scaled by its characteristic range (e.g. $\bar{V} = V/V_{\text{char}}$, $\bar{n} = n/n_{\text{char}}$); the specific scaling constants are provided in Table 2. The nearest opponent in wrapped arc-length contributes a symmetrically normalized block,

$$\boldsymbol{o}_t^{\text{opp}} = \left[\bar{\Delta s},\ \bar{\Delta n},\ \Delta\bar{V}_s,\ \bar{s}_o,\ \bar{n}_o,\ \bar{g},\ \bar{\tau}_{\text{tc}},\ \bar{V}_o\right]^\top, \tag{22}$$

with the wrapped relative progress

$$\Delta s = \big((s_o - s_e + L_{\text{trk}}/2) \bmod L_{\text{trk}}\big) - L_{\text{trk}}/2, \quad \Delta n = n_o - n_e, \tag{23}$$

$\Delta V_s = V_o - V_e$, the Euclidean gap $g = \sqrt{\Delta s^2 + \Delta n^2}$, and the time-to-collision $\tau_{\text{tc}} = \Delta s/\max(-\Delta V_s, \varepsilon_\tau)$ for $\Delta V_s < 0$. Three one-hot closing-speed buckets $\boldsymbol{b}_t \in \{0,1\}^3$, the locked overtake-side indicator $\sigma_t^{\text{lock}} \in \{-1, 0, +1\}$, and the FSM-mode one-hot encoding $\boldsymbol{e}_{m_t} \in \{0,1\}^{|\mathcal{M}|}$ complete the observation vector

$$\boldsymbol{o}_t = \left[\boldsymbol{o}_t^{\text{ego}}, \boldsymbol{o}_t^{\text{opp}}, \boldsymbol{b}_t, \sigma_t^{\text{lock}}, \boldsymbol{e}_{m_t}\right] \in \mathbb{R}^{27}. \tag{24}$$

Neural fixed-track prior The policy feature extractor owns a fixed map of the circuit inside the neural network. A non-trainable lookup table $\mathcal{P} = \{\boldsymbol{\rho}_i\}_{i=0}^{M-1}$ is constructed offline from the baseline raceline before training. Querying this table at arc-length s returns a compact encoding of the track geometry over the next d_p metres,

$$\begin{aligned}\boldsymbol{\rho}(s) = \big[&\bar{\kappa}_0,\ \bar{\kappa}^{\max}_{0:d_1},\ \bar{\kappa}^{\max}_{d_1:d_2},\ \bar{\kappa}^{\max}_{d_2:d_3},\ \bar{\kappa}^{\max}_{d_3:d_p},\ \bar{\kappa}_{\text{mean}},\ \bar{\kappa}_{\text{std}},\\ &\bar{W}^{\text{L}}_{\min},\ \bar{W}^{\text{R}}_{\min},\ \bar{W}^{\text{L}}_0,\ \bar{W}^{\text{R}}_0,\ \Delta\bar{W}_{\min}\big]^\top,\end{aligned} \tag{25}$$

where the first five entries are the local curvature and the peak absolute curvatures in successive d_{win}-metre windows; the remaining entries encode current and bottleneck left/right track half-widths, each normalized by the characteristic track width W_{char}. The actor and critics therefore receive the map-conditioned input

$$\widetilde{\boldsymbol{o}}_t = \left[\boldsymbol{o}_t, \boldsymbol{\rho}(s_e), \boldsymbol{\rho}(s_o)\right] \in \mathbb{R}^{d_o + 2d_\rho}, \tag{26}$$

processed by a feedforward feature network. Critically, the track map is embedded in the neural network as a fixed spatial prior indexed by ego and opponent arc-length; it is not a hand-coded planner cost term.

Table 3: Elementwise bounds of the TQC physical action vector $\boldsymbol{a}_t$.

Entry	$\underline{\boldsymbol{a}}$	$\overline{\boldsymbol{a}}$
L_t^{cap}	1.5	6.0
R_t^{cap}	1.5	6.0
d_{safe}	0.2	0.8
V_{cap}	15	90
g_{chase}	18	24
g_{ot}	16	24
g_{abort}	28	45
b_{lat}	-1.5	1.5
w_{safe}	1	5
q_{side}	-1	1

Action The TQC actor outputs a bounded vector $\boldsymbol{u}_t \in [-1,1]^{d_a}$, which is affine-mapped elementwise to the physical action

$$\boldsymbol{a}_t = \underline{\boldsymbol{a}} + \tfrac{1}{2}(\boldsymbol{u}_t + \mathbf{1}) \odot (\overline{\boldsymbol{a}} - \underline{\boldsymbol{a}}), \tag{27}$$

where the action vector

$$\begin{aligned}\boldsymbol{a}_t = \big[&L_t^{\text{cap}},\, R_t^{\text{cap}},\, d_{\text{safe}},\, V_{\text{cap}},\, g_{\text{chase}},\, g_{\text{ot}},\\ &g_{\text{abort}},\, b_{\text{lat}},\, w_{\text{safe}},\, q_{\text{side}}\big]^\top\end{aligned} \tag{28}$$

comprises the left/right corridor caps $(L_t^{\text{cap}}, R_t^{\text{cap}})$, the safety margin d_{safe}, the velocity cap V_{cap}, the FSM hysteresis thresholds $(g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}})$, a lateral bias b_{lat}, a corridor-width bonus w_{safe}, and a side-selection score $q_{\text{side}} \in [-1,1]$ whose sign determines the overtake direction ($q_{\text{side}} \geq 0$: left; $q_{\text{side}} < 0$: right). The side decision is locked for the duration of one duel. The elementwise bounds $\underline{\boldsymbol{a}}, \overline{\boldsymbol{a}}$ are provided in Table 3. After affine mapping, consistency clips

$$\begin{aligned} g_{\text{chase}} &\geq g_{\text{ot}} - \varepsilon_{co}, \quad g_{\text{ot}} \geq g_{\text{chase}} - \varepsilon_{oc},\\ g_{\text{abort}} &\geq g_{\text{chase}} + \varepsilon_{ca}, \quad L_t^{\text{cap}} + R_t^{\text{cap}} \geq \varepsilon_{\text{corr}}. \end{aligned} \tag{29}$$

are enforced with small positive offsets $(\varepsilon_{co}, \varepsilon_{oc}, \varepsilon_{ca}, \varepsilon_{\text{corr}})$ to maintain a non-empty corridor and a well-ordered FSM throughout training.

Finite-state machine The discrete tactical mode $m_t \in \mathcal{M} = \{$Raceline, Shadow, Overtake, Hold, Defend$\}$ is selected by a hysteretic FSM. Let g denote the longitudinal gap to the leading opponent and g_{rear} the gap to the closest trailing opponent. The FSM transition logic is

$$m_{t+1} = \begin{cases} \textsc{Overtake} & g \leq g_{\text{ot}} \,\wedge\, \textit{ready} \,\wedge\, \bar{c} \\ \textsc{Shadow} & g_{\text{ot}} < g \leq g_{\text{chase}} \\ \textsc{Hold} & m_t = \textsc{Overtake} \wedge \textit{abort} \\ \textsc{Defend} & g > g_{\text{def}}^{\text{front}} \wedge 0 < g_{\text{rear}} < g_{\text{def}}^{\text{rear}} \wedge \bar{c} \\ \textsc{Raceline} & g > g_{\text{chase}}, \end{cases} \tag{30}$$

where $\bar{c}$ indicates that the post-overtake immunity window has expired. The abort criterion is $(g > g_{\text{abort}}) \vee (\Delta n_{\min} < \Delta n_{\text{safe}} \wedge g < g_{\text{interlock}})$, which guards against late-corner divergence and lateral interlock with the opponent. The thresholds $(g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}})$ are the RL-modulated hysteresis parameters from $\boldsymbol{a}_t$, while $(g_{\text{def}}^{\text{front}}, g_{\text{def}}^{\text{rear}}, \Delta n_{\text{safe}}, g_{\text{interlock}})$ are fixed constants whose values are reported in Table 4.

Reward The per-step reward used to train the TQC policy is the weighted sum

$$r_t = \sum_j w_j\, \phi_j(t), \tag{31}$$

where the feature set and weights are detailed in Table 5. Key features include: an overtake success detector $\phi_{\text{ot}} = ⊮[\Delta s_{t-1} > \Delta s_{\text{ot}}^+ \wedge \Delta s_t < -\Delta s_{\text{ot}}^-]$ with hysteresis margins $(\Delta s_{\text{ot}}^+, \Delta s_{\text{ot}}^-)$; an approach incentive $\phi_{\text{app}} = ⊮[g < g_{\text{app}} \wedge \dot{g} < 0]\,/(1+g)$ that encourages closing on a distant leader; and a

Table 4: Fixed FSM constants (not modulated by RL).

Symbol	Value
$g_{\mathrm{def}}^{\mathrm{front}}$	25 m
$g_{\mathrm{def}}^{\mathrm{rear}}$	20 m
Δn_{safe}	2.5 m
$g_{\mathrm{interlock}}$	6 m
Post-overtake immunity	30 s

Table 5: Reward weights of the TQC tactical policy.

Feature	Weight w_j	Meaning
ϕ_{ot}	+30	Overtake sign-flip
ϕ_{ovb}	−20	Overtaken by opponent
ϕ_{col}	−100	Collision (terminal)
$\phi_{\mathrm{lat}} = e_{\mathrm{lat}}^2$	−2	Lateral tracking error
ϕ_{crush}	−5	$L^{\mathrm{cap}} + R^{\mathrm{cap}} < 0.8$
ϕ_{app}	+3	Approach bonus
$\phi_{\mathrm{sm}} = \lVert \Delta \boldsymbol{a}_t \rVert^2$	−0.1	Action smoothness
ϕ_{gs}	−0.5	Geometric smoothness
ϕ_{sbs}	+0.5	Side-by-side combat
ϕ_{wide}	+0.25	Wide-corridor incentive

geometric smoothness penalty $\phi_{\mathrm{gs}} = (\Delta L_t^{\mathrm{cap}})^2 + (\Delta R_t^{\mathrm{cap}})^2 + (\Delta b_{\mathrm{lat},t})^2$ that suppresses corridor jitter between consecutive steps.

The policy is optimized with Truncated Quantile Critics. Each critic represents a distributional action value by N_q quantile atoms $\{z_{\phi_i}^j(\widetilde{\boldsymbol{o}}, \boldsymbol{u})\}_{j=1}^{N_q}$; the truncated target distribution discards the largest atoms before Bellman backup,

$$\mathcal{T}_{\mathrm{trunc}} = \mathrm{sort}\Big(\{z_{\bar{\phi}_i}^j(\widetilde{\boldsymbol{o}}', \boldsymbol{u}') - \alpha \log \pi_\theta(\boldsymbol{u}'|\widetilde{\boldsymbol{o}}')\}_{i,j}\Big)_{1:K}, \tag{32}$$

where $K < N_q$ is the number of retained quantile atoms. The critic minimizes the quantile Huber loss

$$J_Q(\phi) = \sum_{i,j} \mathbb{E}_{\mathcal{D},\, y \in r + \gamma \mathcal{T}_{\mathrm{trunc}}} \left[\rho_{\hat{\tau}_j}^{\kappa}\Big(y - z_{\phi_i}^j(\widetilde{\boldsymbol{o}}, \boldsymbol{u})\Big)\right], \tag{33}$$

while the actor maximizes the mean truncated value with an entropy regularizer,

$$J_\pi(\theta) = \mathbb{E}_{\widetilde{\boldsymbol{o}} \sim \mathcal{D},\, \boldsymbol{u} \sim \pi_\theta}\left[\alpha \log \pi_\theta(\boldsymbol{u}|\widetilde{\boldsymbol{o}}) - \frac{1}{K} \sum_{z \in \mathcal{T}_{\mathrm{trunc}}(\widetilde{\boldsymbol{o}}, \boldsymbol{u})} z\right]. \tag{34}$$

The temperature α is adjusted automatically to maintain a target entropy. Training hyperparameters are summarized in Table 6.

One training episode consists of a single overtake duel: an NPC opponent is spawned ahead of the ego, and the episode terminates on successful overtake, collision, off-track excursion, loss of V2V communication, or a timeout. All quantities appearing in the simulation analysis (Sec. 7) use the notation established in this section: $\widetilde{\boldsymbol{o}}_t$ for the map-conditioned observation, $\boldsymbol{\rho}(s)$ for the track prior, $\boldsymbol{a}_t$ and q_{side} for the RL action, and m_t for the FSM mode that generates the corridor.

3.2 Heading-Aware Corridor Carving

The corridor carver is a deterministic function

$$\begin{aligned} \mathcal{F}_{\mathrm{carve}} : (\boldsymbol{o}_t, \boldsymbol{a}_t, m_t, \{\boldsymbol{o}_t^{(j)}\}_{j=1}^{N_{\mathrm{opp}}}) \\ \mapsto \{(s_k, L_{\min,k}, L_{\max,k})\}_{k=0}^{N_{\mathrm{sta}}-1} \end{aligned} \tag{35}$$

that converts opponent states and the FSM mode into a corridor of N_{sta} stations with uniform spacing Δs_{sta} over the planning horizon d_p. All numerical constants referenced below are collected in Table 7.

Table 6: TQC training hyperparameters.

Hyperparameter	Value
Learning rate	3×10^{-4}
Replay buffer size	10^5
Batch size	256
Discount factor γ	0.99
Polyak τ	5×10^{-3}
Feature network	$51 \to 256 \to 256$ (ReLU)
Policy/Critic MLP	$[256, 256]$
Quantile atoms N_q	25
Truncated atoms K	20

Heading-aware projection For opponent j with Frenet heading error $\chi^{(j)}$ and rectangular footprint of length L_v and width W_v, the half-extents in the Frenet (s, n) axes are

$$\begin{aligned} \rho_n^{(j)} &= \tfrac{1}{2}\big(L_v|\sin\chi^{(j)}| + W_v|\cos\chi^{(j)}|\big), \\ \rho_s^{(j)} &= \tfrac{1}{2}\big(L_v|\cos\chi^{(j)}| + W_v|\sin\chi^{(j)}|\big). \end{aligned} \tag{36}$$

The lateral exclusion radius around opponent j is then

$$\rho_{\mathrm{exc}}^{(j)} = \rho_n^{(j)} + \tfrac{W_v}{2} + d_{\mathrm{safe}}, \tag{37}$$

where d_{safe} is the RL-modulated safety margin from $\boldsymbol{a}_t$.

Spatial fading and startup protection Opponent influence is attenuated by a smooth fading kernel over distance,

$$\phi_{\mathrm{fade}}(\Delta s) = \begin{cases} \cos^{p_{\mathrm{fade}}}\!\Big(\frac{\pi}{2}\frac{|\Delta s|}{S_{\mathrm{fade}}}\Big), & |\Delta s| < S_{\mathrm{fade}}, \\ 0, & \text{otherwise}, \end{cases} \tag{38}$$

with characteristic range S_{fade}. A linear startup ramp $\phi_{\mathrm{sr}}(k)$ prevents corridor collapse in the first few stations immediately ahead of the ego, where the planner has insufficient spatial authority to react.

Mode-conditioned corridor update Let $L_{\max,k}^{(0)}, L_{\min,k}^{(0)}$ be the geometric track-edge half-widths at station k. In Overtake mode with committed pass side $\sigma \in \{+1, -1\}$, the carver tightens the opponent-side bound,

$$L_{\sigma,k} \leftarrow \sigma\big(\sigma\, n_{o,k} + \rho_{\mathrm{exc}}^{(j)}\, \phi_{\mathrm{fade}}\, \phi_{\mathrm{sr}}\, \phi_{\mathrm{rmp}}\big), \tag{39}$$

where $n_{o,k}$ is the predicted opponent lateral position at s_k and $\phi_{\mathrm{rmp}}(k)$ is a distance-dependent blending ramp. In Shadow mode the corridor is constructed as an asymmetric funnel that widens from the ego toward the pass side; the funnel width at convergence and the near-ego bonus are controlled by the parameters $(W_{\mathrm{fun}}, W_{\mathrm{near}}, \lambda_{\mathrm{side}})$, with λ_{side} governing the pass-side allocation fraction. Across all modes a minimum corridor width $L_{\min}^{\mathrm{corr}}$ is enforced, and the resulting bound profile is temporally smoothed by an exponential moving average with coefficient $\beta \in (0, 1)$,

$$\big[L_{\min,k}, L_{\max,k}\big]_t \;\leftarrow\; \beta\,\big[\,\cdot\,\big]_t + (1-\beta)\big[\,\cdot\,\big]_{t-1}. \tag{40}$$

RL bias blending The RL action also emits a terminal lateral bias b_{lat} and global caps $(L_t^{\mathrm{cap}}, R_t^{\mathrm{cap}})$. These are blended into the carver output through a quadratic weight w_k that vanishes at the ego ($k = 0$) and reaches unity at a short look-ahead index k_{blend},

$$w_k = (k/k_{\mathrm{blend}})^2\, \nVdash[k < k_{\mathrm{blend}}] + \nVdash[k \geq k_{\mathrm{blend}}], \tag{41}$$

$$L_{\max,k} \leftarrow (1 - w_k)L_{\max,k} + w_k \min\big(L_{\max,k},\, b_{\mathrm{lat}} + L_t^{\mathrm{cap}}\big), \tag{42}$$

$$L_{\min,k} \leftarrow (1 - w_k)L_{\min,k} + w_k \max\big(L_{\min,k},\, b_{\mathrm{lat}} - R_t^{\mathrm{cap}}\big). \tag{43}$$

This blending is the mechanism through which the RL action directly shapes the planner corridor; the safety margin and side decision enter upstream through the carver, and all three are ultimately realized as $\mathcal{C}_k$.

Table 7: Corridor-carving constants.

Symbol	Value
d_p	300 m
N_{sta}	150
Δs_{sta}	2 m
p_{fade}	3
S_{fade}	50 m
W_{fun}	2.5 m (per side)
W_{near}	5 m
λ_{side}	0.6
$L^{\text{corr}}_{\text{min}}$	3 m
β (EMA)	0.2
k_{blend}	9
d_{rmp}	40 m
k_{sr}	15

4 Local Planning Layer

Given the corridor $\mathcal{C} = \{(s_k, L_{\min,k}, L_{\max,k})\}$ from the tactical layer, the planner produces a curvature-smooth lateral profile $l(s)$ inside $\mathcal{C}$ via a SOCP, and a maximum-grip speed profile $v(s)$ via a forward–backward sweep on the GG super-ellipse. Both solvers are exact and convex on their respective subproblems; convexity is preserved by sequential linearization of the only nonlinearity in (17).

4.1 SOCP Lateral Path Planning

Decision variable At each of N stations $k = 0, \dots, N-1$ along the look-ahead horizon d_p, with uniform spacing $\Delta s_k = d_p/(N-1)$, the planner instantiates four scalars,

$$\boldsymbol{y}_k = \left[\, l_k,\ \alpha_k,\ \zeta_k,\ \sigma_k \,\right]^\top \in \mathbb{R}^4, \tag{44}$$

namely lateral offset l_k, heading deviation α_k, curvature slack ζ_k, and the second-order cone witness σ_k. The stacked decision vector is $\boldsymbol{y} = [\boldsymbol{y}_0^\top, \dots, \boldsymbol{y}_{N-1}^\top]^\top \in \mathbb{R}^{4N}$. For numerical conditioning, all quantities are non-dimensionalized by a characteristic length r_{sc} and the derived speed $v_{\text{sc}} = \sqrt{r_{\text{sc}}\, g}$.

Cost The cost penalizes only the cone witness,

$$J_{\text{SOCP}}(\boldsymbol{y}) = \sum_{k=0}^{N-1} w_\sigma\, \sigma_k, \tag{45}$$

which, together with the cone constraint defined below, implements an L_2 penalty on the curvature slack. Path smoothness is induced indirectly by the dynamic equality: $l_{k+1} - l_k$ is slaved to $\alpha_k \Delta s_k$, which is in turn coupled to σ_k through (17); minimizing σ_k therefore minimizes a quadratic surrogate of $l''(s)$.

Equality constraints Initial and (mode-dependent) terminal conditions pin the boundary,

$$l_0 = l_{\text{ego}},\ \ \alpha_0 = \alpha_{\text{ego}}, \qquad l_{N-1} = l_f^\star,\ \ \alpha_{N-1} = \alpha_f^\star, \tag{46}$$

where the terminal target is selected as

$$l_f^\star = \begin{cases} l_{\text{raceline}}(s_{N-1}) & \text{race mode, in-corridor,} \\ \frac{1}{2}(L_{\max,N-1} + L_{\min,N-1}) & \text{tactical/out-corridor.} \end{cases} \tag{47}$$

The Frenet dynamics (17) are discretized by trapezoidal integration; for $i = 0, \dots, N-2$,

$$\begin{aligned} &\left(\boldsymbol{I} + \tfrac{\Delta s_i}{2}\boldsymbol{A}_{1,i}\right)\boldsymbol{\xi}_i + \tfrac{\Delta s_i}{2}\boldsymbol{B}_{1,i}\boldsymbol{\eta}_i \\ &\qquad - \left(\boldsymbol{I} - \tfrac{\Delta s_i}{2}\boldsymbol{A}_{2,i}\right)\boldsymbol{\xi}_{i+1} + \tfrac{\Delta s_i}{2}\boldsymbol{B}_{2,i}\boldsymbol{\eta}_{i+1} \\ &\qquad\qquad = -\tfrac{\Delta s_i}{2}\left(\boldsymbol{b}_{1,i} + \boldsymbol{b}_{2,i}\right), \end{aligned} \tag{48}$$

with $\boldsymbol{\xi}_i = [l_i, \alpha_i]^\top$, $\boldsymbol{\eta}_i = [\zeta_i, \sigma_i]^\top$ and $(\boldsymbol{A}_{j,i}, \boldsymbol{B}_{j,i}, \boldsymbol{b}_{j,i})$ the linearization (17) at endpoints $j \in \{1, 2\}$ of segment i.

Linear inequality constraints (corridor) For each station k, the corridor injected by the tactical layer is mapped into two affine half-planes,

$$L_{\min,k} + d_{\text{veh}} \le l_k \le L_{\max,k} - d_{\text{veh}}, \tag{49}$$

where d_{veh} accounts for the half-vehicle width plus a static safety margin. In race mode without external corridor injection, $L_{\min,k}, L_{\max,k}$ default to the geometric track edges reduced by the on-line obstacle clearance.

Second-order cone constraint The curvature slack and its epigraph variable are bound through a single 3-dimensional cone per station,

$$\left\| \left[2\zeta_k,\ \sigma_k - 1 \right]^\top \right\|_2 \le \sigma_k + 1, \quad k = 0, \ldots, N-1, \tag{50}$$

which is equivalent to $\zeta_k^2 \le \sigma_k$ for $\sigma_k \ge 0$. Combined with the linear cost in (45), the constraint implements a convex quadratic epigraph penalty on curvature slack without introducing a dense Hessian in the conic program.

Compiled SOCP Defining $\boldsymbol{c} \in \mathbb{R}^{4N}$ with $\boldsymbol{c}_{4k+3} = w_\sigma$ and zero elsewhere, $\boldsymbol{G} \in \mathbb{R}^{(2N+3N)\times 4N}$ stacking the corridor and cone rows, and $\boldsymbol{A} \in \mathbb{R}^{(2+2(N-1)+2)\times 4N}$ stacking (46)–(48), the planner solves

$$\begin{aligned} \boldsymbol{y}^\star = \arg\min_{\boldsymbol{y}\in\mathbb{R}^{4N}} \ & \boldsymbol{c}^\top \boldsymbol{y} \\ \text{s.t.}\ & \boldsymbol{A}\boldsymbol{y} = \boldsymbol{b}, \\ & \boldsymbol{G}\boldsymbol{y} \preceq_{\mathcal{K}} \boldsymbol{h}, \end{aligned} \tag{51}$$

where $\mathcal{K} = \mathbb{R}_+^{2N} \times \mathcal{Q}_3^N$ is the product of the linear inequality cone and N second-order cones of dimension three. Both $\boldsymbol{G}$ and $\boldsymbol{A}$ are assembled directly in compressed column storage.

Sequential linearization The matrices in (17) depend on the operating point $(l_i, \alpha_i, \kappa_i)$. We resolve this by sequential linearization: at iteration j, (51) is solved with $(l_i^{(j)}, \alpha_i^{(j)})$ as the linearization point; the new iterate $(l_i^{(j+1)}, \alpha_i^{(j+1)})$ is read from $\boldsymbol{y}^\star$; convergence is declared when

$$\max_i |l_i^{(j+1)} - l_i^{(j)}| < \varepsilon_l, \qquad \max_i |\alpha_i^{(j+1)} - \alpha_i^{(j)}| < \varepsilon_\alpha, \tag{52}$$

or the iteration cap $J_{\max}$ is reached. In practice, two to three iterations suffice once the iterate is warm-started from the previous planning cycle's solution.

4.2 Forward–Backward GG-Coupled Velocity Profiling

The path $l^\star(s)$ obtained from (51) is converted into a 2D curve $\boldsymbol{p}(s) = \boldsymbol{r}(s) + l^\star(s)\boldsymbol{n}(s)$ whose curvature $\kappa^\star(s)$ is recomputed analytically. The FB sweep then assigns a speed profile $v(s)$ that maximizes lap pace while respecting the super-elliptic GG envelope (18) and the speed-dependent grip caps $a_y^{\max}(v)$, $a_x^{\max}(v)$.

Curvature ceiling At each station the maximum stationary speed is bounded by lateral grip, $v_{\text{ce}}^2(s) = a_y^{\max}(v_{\text{ce}})/|\kappa^\star(s)|$. The implicit dependence of $a_y^{\max}$ on v is resolved by fixed-point iteration,

$$v_{\text{ce}}^{(j+1)} = \sqrt{a_y^{\max}\big(v_{\text{ce}}^{(j)}\big)/|\kappa^\star(s)|}, \tag{53}$$

with stopping tolerance ε_v, typically converging in fewer than five iterations. The result is the station-wise curvature-ceiling array $V_{\text{ce}}[k]$.

Forward pass (acceleration) Initialized at $V_f[0] = \min(v_{\text{ego}}, V_{\text{ce}}[0])$, the forward sweep integrates from $k = 0$ to $N-2$ as

$$a_{y,k} = \kappa_k^\star\, V_f[k]^2, \tag{54}$$

$$\begin{aligned} a_{x,k}^{\text{avail}} &= a_x^{\max}(V_f[k]) \\ &\quad \times \left[1 - (|a_{y,k}|/a_y^{\max}(V_f[k]))^p\right]^{1/p}, \end{aligned} \tag{55}$$

$$a_{x,k}^{\text{net}} = \min\!\big(a_{x,k}^{\text{avail}}, a_{x,k}^{\text{engine}}\big) + a_{\text{drag}}(V_f[k]), \tag{56}$$

$$V_f[k+1] = \min\Big(\sqrt{V_f[k]^2 + 2\Delta s_k\, a^{\text{net}}_{x,k}},\ V_{\text{ce}}[k+1]\Big), \tag{57}$$

where $a^{\text{engine}}_{x,k}$ is the powertrain ceiling read from the PTP-on/off lookup tables and $a_{\text{drag}}(\cdot)$ is the drag model of (20). This sweep encodes the maximum-grip acceleration policy under combined slip.

Backward pass (braking) Terminated at $V_b[N-1] = \min(v_f, V_{\text{ce}}[N-1])$, the backward sweep runs from $k = N-2$ down to 0,

$$\tilde{a}_{y,k+1} = \kappa^\star_{k+1}\, V_b[k+1]^2, \tag{58}$$

$$\begin{aligned}\tilde{a}^{\text{brk}}_{x,k+1} &= a^{\max}_x(V_b[k+1]) \\ &\quad \times \big[1 - (|\tilde{a}_{y,k+1}|/a^{\max}_y(V_b[k+1]))^p\big]^{1/p},\end{aligned} \tag{59}$$

$$\tilde{a}^{\text{tot}}_{x,k+1} = \tilde{a}^{\text{brk}}_{x,k+1} - a_{\text{drag}}(V_b[k+1]), \tag{60}$$

$$V_b[k] = \min\Big(\sqrt{V_b[k+1]^2 + 2\Delta s_k\, \tilde{a}^{\text{tot}}_{x,k+1}},\ V_{\text{ce}}[k]\Big). \tag{61}$$

Drag enters with opposite sign to (56) because it *assists* braking; this sign discipline is essential to match on-track coastdown traces above 50 m/s.

Profile fusion and acceleration extraction The final speed profile is the pointwise minimum,

$$V[k] = \min\big\{V_f[k],\ V_b[k],\ V_{\text{ce}}[k]\big\}, \tag{62}$$

clipped to $[v_{\min}, v_{\max}]$ for numerical stability. Tangential and normal accelerations follow from energy conservation,

$$a_{T,k} = \frac{V[k+1]^2 - V[k]^2}{2\Delta s_k}, \qquad a_{N,k} = \kappa^\star_k V[k]^2. \tag{63}$$

A runtime aggressiveness factor $\eta \in (0,1]$ and an upstream velocity cap v_{rc} are applied as the final scaling and clipping,

$$V_{\text{out}}[k] = \min\big(\eta\, V[k],\ v_{\text{rc}}\big). \tag{64}$$

The output trajectory $\{(x_k, y_k, \psi_k, \kappa_k, V_{\text{out}}[k], t_k)\}$, with time stamps $t_{k+1} = t_k + \Delta s_k / V_{\text{out}}[k]$, is resampled at the controller cadence Δt_{out} to produce a reference of N_{out} points.

5 Control Layer: MPCC

The controller closes the loop on the planner output with a Model Predictive Contouring Controller (MPCC) that exploits the dynamic model (10) and explicitly trades tracking accuracy against arc-length progress through a single positively-weighted terminal term.

5.1 Stage Cost and Progress Maximization

At each control cycle of period $T_c = 0.05$ s, the controller solves a finite-horizon nonlinear program over $N_c = 15$ stages, with a horizon of $T_c N_c = 0.75$ s. The stage cost couples a weighted lateral, yaw and speed tracking error with input regularization,

$$\begin{aligned}L_k(\boldsymbol{x}_k, \boldsymbol{u}_k, \zeta_k) &= q_l e^2_{l,k} + q_\psi e^2_{\psi,k} + q_v e^2_{v,k} \\ &\quad + R_a a_k^2 + R_{\dot\delta} \dot\delta_k^2 + w_\zeta \zeta_k^2,\end{aligned} \tag{65}$$

where the contouring/lag decomposition is

$$\begin{aligned}e_{l,k} &= -\sin\psi_{r,k}(p_{x,k} - x_{r,k}) \\ &\quad + \cos\psi_{r,k}(p_{y,k} - y_{r,k}),\end{aligned} \tag{66}$$

$$e_{\psi,k} = \psi_k - \psi_{r,k}, \qquad e_{v,k} = v_{x,k} - v_{r,k}, \tag{67}$$

with $(x_{r,k}, y_{r,k}, \psi_{r,k}, v_{r,k})$ the reference quadruple from the planner expressed in the body frame at $k = 0$ to avoid the ill-conditioning of large global positions. The reference speed entering $e_{v,k}$ is clamped per stage,

$$v_{r,k} \leftarrow \text{clamp}\big(v_{r,k},\ v_x - \Delta v_{\text{lo}},\ v_x + \Delta v_{\text{hi}}\big), \tag{68}$$

which prevents the velocity-tracking term from dominating during sharp deceleration phases where the planner naturally lags behind the actual vehicle speed. The asymmetric bounds $(\Delta v_{\text{lo}}, \Delta v_{\text{hi}})$

Table 8: MPCC stage and terminal cost weights, and bounds.

Symbol	Value	Description
q_l	50	Lateral error
q_ψ	5	Yaw error
q_v	1	Speed error
R_a	0.05	Acceleration regularization
$R_{\dot\delta}$	30	Steering-rate regularization
q_{prog}	0.5	Progress reward (terminal)
w_ζ	1000	Friction slack penalty
N_c, T_c	15, 0.05 s	Horizon length, sample time

are chosen such that the controller is permitted to overshoot the reference but penalized for undershooting.

The terminal cost is the unique asymmetric term,

$$\Phi(\boldsymbol{x}_{N_c}) = -q_{\text{prog}}\, s_{N_c}, \qquad q_{\text{prog}} = 0.5, \tag{69}$$

which rewards arc-length progress along the prediction window. Because s is integrated as $\dot s = v_x$ in (1), maximizing s_{N_c} is equivalent to maximizing the path-tangent component of average speed over the horizon, weighted toward the end of the horizon. Combined with the friction-circle constraint introduced below, this term is the mechanism by which the controller produces the canonical entry-deceleration / apex-grip / exit-acceleration pattern without an explicit corner-by-corner reference shaping.

The MPCC weights are summarized in Table 8; they were tuned once on the global raceline and held constant across all races.

5.2 Friction-Circle Constrained NLP and Real-Time Solution

Constraints Equality constraints enforce the dynamics and the initial state,

$$\boldsymbol{x}_{k+1} = f_d(\boldsymbol{x}_k, \boldsymbol{u}_k),\ k = 0, \ldots, N_c - 1; \quad \boldsymbol{x}_0 = \boldsymbol{x}_{\text{init}}. \tag{70}$$

Box constraints on inputs and states are applied at every stage,

$$\underline{a} \le a_k \le \overline{a}, \quad -\dot\delta_{\max} \le \dot\delta_k \le \dot\delta_{\max}, \tag{71}$$

$$0 \le v_{x,k} \le v_{x,\max}, \quad -\delta_{\max} \le \delta_k \le \delta_{\max}. \tag{72}$$

Combined slip on the rear axle is enforced as a single soft second-order cone per stage,

$$\big(m\, a_k\big)^2 + \big(C_r\, \alpha_{r,k}\big)^2 \le \big(\mu F_{z,r} + \zeta_k\big)^2, \quad \zeta_k \ge 0, \tag{73}$$

with the rear normal load determined by static weight distribution,

$$F_{z,r} = \frac{mg\, l_f}{l_f + l_r}. \tag{74}$$

The slack ζ_k in (73) is the same variable that appears in L_k through $w_\zeta \zeta_k^2$; this places a quadratic price on exceeding the friction envelope and is the primary mechanism by which the controller stays inside (18) without rendering the NLP infeasible during transient over-grip events.

Compiled NLP Collecting the decision variables $(\boldsymbol{X}, \boldsymbol{U}, \boldsymbol{\zeta}) \in \mathbb{R}^{8(N_c+1)} \times \mathbb{R}^{2N_c} \times \mathbb{R}_+^{N_c}$, the controller solves

$$\begin{aligned} \min_{\boldsymbol{X},\boldsymbol{U},\boldsymbol{\zeta}} \ & \sum_{k=0}^{N_c-1} L_k(\boldsymbol{x}_k, \boldsymbol{u}_k, \zeta_k) + \Phi(\boldsymbol{x}_{N_c}) \\ \text{s.t. } & (70)\text{–}(73) \text{ hold.} \end{aligned} \tag{75}$$

The NLP is solved by a primal–dual interior-point method with adaptive barrier strategy, warm-started from the previous cycle's solution shifted by one stage. An iteration limit and a wall-time budget are imposed to guarantee deterministic returns within the control period. The steering-angle and acceleration commands are reconstructed from the optimal solution as

$$\delta_0^\star = \delta_{\text{prev}} + T_c\, \dot\delta_0^\star, \qquad a_0^\star = a^\star. \tag{76}$$

Table 9: Closed-loop metrics for the two reported overtake simulations.

Case	Δt_{pass} [s]	$\max V$ [m/s]	RMS(e_l) [m]	RMS(e_v) [m/s]	$\Delta\Gamma$ [-]
C2	1.13	73.10	0.02	0.26	1.38
C5	8.63	65.83	0.05	1.82	1.75

Real-time monitoring and fallback The MPCC solution is accepted only when (i) the solver returns a success status, (ii) the commanded acceleration and steering angle lie within predefined safety bounds $|a_0^\star| \le a_{\max}$ and $|\delta_0^\star| \le \delta_{\max}$, and (iii) the body speed is above the minimum threshold for which the linearized tire model in (5) remains valid. If any check fails, the controller reverts to a legacy lateral MPC plus longitudinal PI cascade for the current cycle and reattempts MPCC on the next cycle.

6 Real-Time Integration

The hierarchy is implemented as an acyclic cascade with explicit data contracts between adjacent layers. At tactical time t, the policy and FSM publish a guidance tuple

$$\begin{aligned} \mathcal{G}_t &= \big(m_t,\ V_{\text{cap},t},\ \mathcal{C}_t,\ \pi_t\big), \\ \mathcal{C}_t &= \{(s_k, L_{\min,k}, L_{\max,k})\}_{k=0}^{N_{\text{sta}}-1}, \end{aligned} \tag{77}$$

where m_t selects the terminal condition in (47), $V_{\text{cap},t}$ clips the FB speed profile through (64), $\mathcal{C}_t$ defines the lateral half-plane constraints (49), and π_t stores race-control permissions such as point-to-point passing availability. The planner holds the last valid guidance tuple if a tactical update is delayed; thus the feasible set evolves piecewise smoothly rather than discontinuously.

The planner converts $\mathcal{G}_t$ into a time-parameterized reference

$$\mathcal{R}_t = \{(x_k, y_k, \psi_k, \kappa_k, V_k, \tau_k)\}_{k=0}^{N_{\text{out}}-1}, \tag{78}$$

after the SOCP path solve and GG-constrained velocity sweep. The MPCC controller transforms $\mathcal{R}_t$ into the local body frame at each control cycle, interpolates the reference on its prediction grid, and solves (75). This design leaves only two forms of authority flowing downstream: the tactical layer may restrict the feasible set and speed cap, and the planner may update the reference trajectory. Neither layer can overwrite controller constraints, tire limits, or actuator bounds.

The temporal hierarchy can therefore be summarized as

$$\underbrace{\text{strategy}}_{\text{tactical FSM tenure}} \succ \underbrace{\text{corridor}}_{\text{tactical/planner I/O}} \succ \underbrace{\text{MPCC}}_{\text{control loop}}, \tag{79}$$

with each upper layer setting the admissible set of the layer below. This is the principal systems property used in the experimental analysis: tactical success is judged by whether the learned corridor is executable by the unchanged planner and controller, not by whether a learned controller happens to survive a particular duel.

7 Simulation Results

This section reports the two closed-loop overtake trials used to validate the proposed hierarchy. Both runs use the frozen tactical policy, the same SOCP planner, and the same MPCC controller. No online learning, case-specific retuning, or controller-weight change is applied.

7.1 Protocol and Metrics

The simulations use the Yas Marina North layout with one NPC opponent initialized approximately 60 m ahead of the ego vehicle. A run terminates at the first successful pass ($\Delta s < 0$), collision, off-track excursion, communication loss, or timeout. Table 9 lists the reduced metrics for the two reported cases.

The two cases are complementary. Case 2 is the short fast-closure overtake, while Case 5 is the longer side-by-side maneuver. Both enter Shadow early, commit to Overtake, and finish with positive tactical gain.

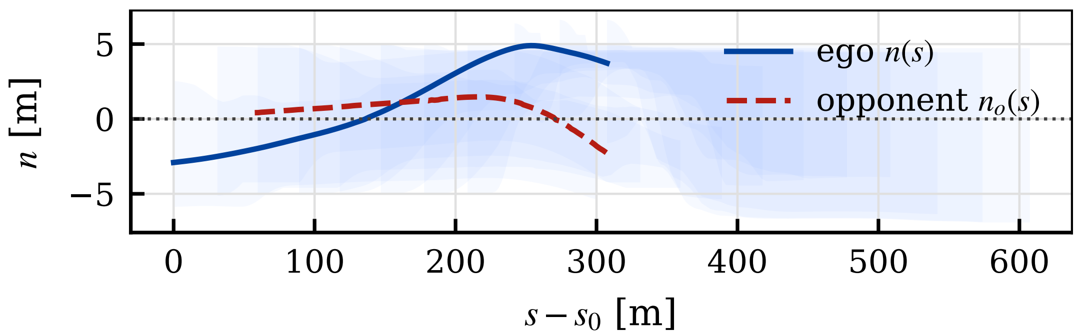


Figure 1: Case 2 Frenet path and tactical corridor. The ego trace remains inside the feasible corridor generated by the tactical layer.

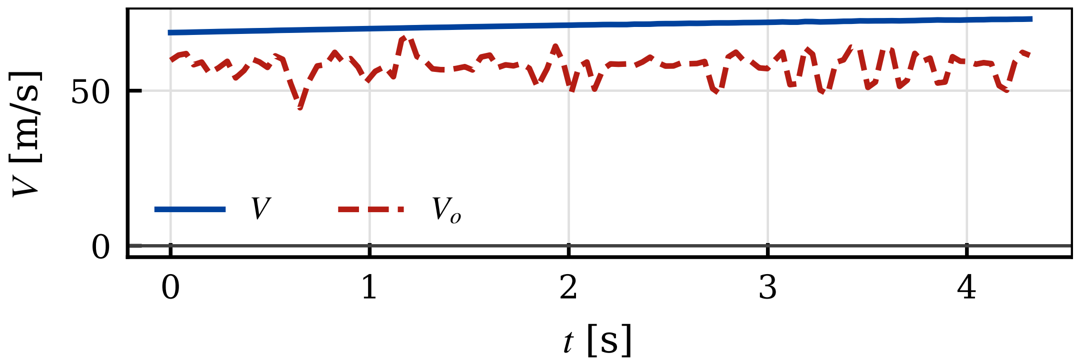


Figure 2: Case 2 speed profile.

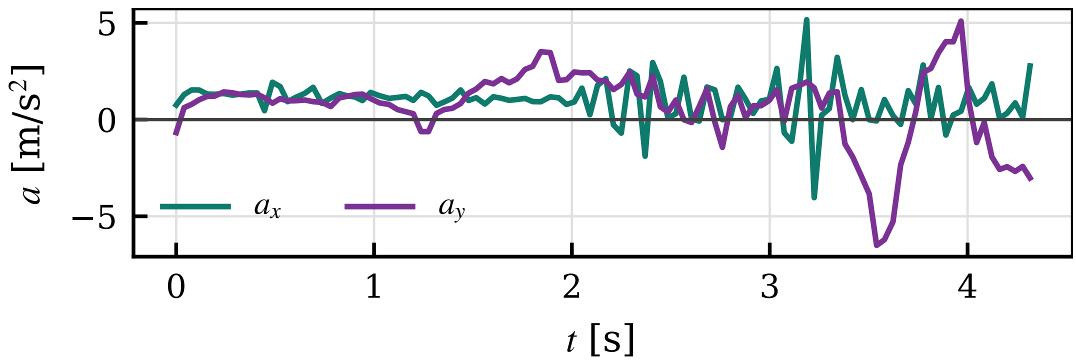


Figure 3: Case 2 longitudinal and lateral acceleration.

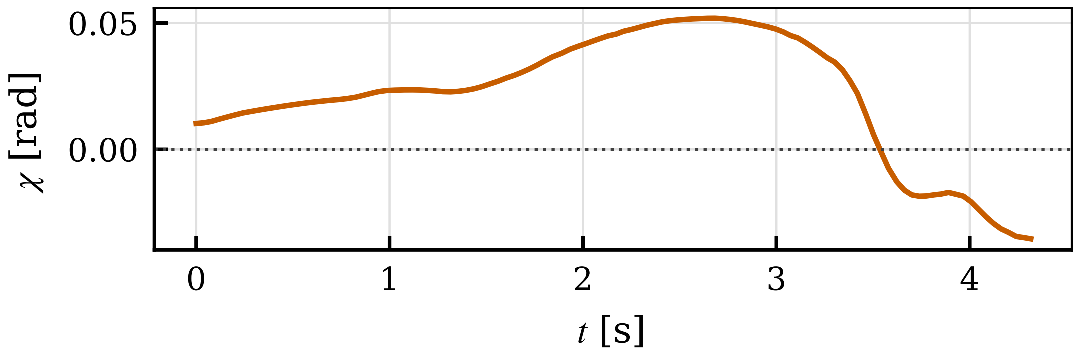


Figure 4: Case 2 heading error.

7.2 Case 2: Short Fast-Closure Pass

This case completes the overtake with $\min \Delta s = -0.48\,\mathrm{m}$, $\Delta t_{\mathrm{pass}} = 1.13\,\mathrm{s}$, $\max V = 73.10\,\mathrm{m/s}$, $\mathrm{RMS}(e_l) = 0.02\,\mathrm{m}$, $\mathrm{RMS}(e_v) = 0.26\,\mathrm{m/s}$, and tactical gain $\Delta\Gamma = 1.38$.

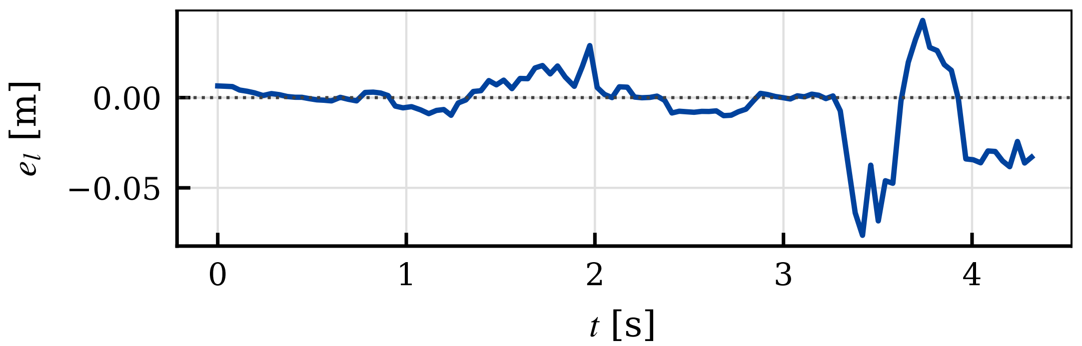


Figure 5: Case 2 lateral tracking error.

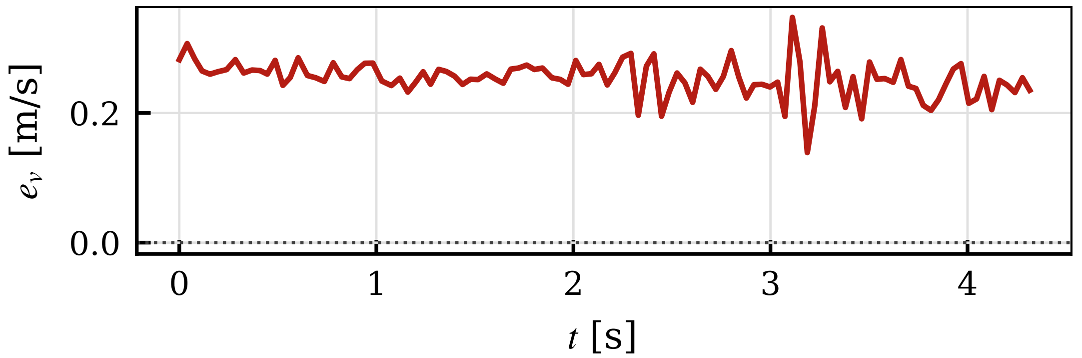


Figure 6: Case 2 speed tracking error.

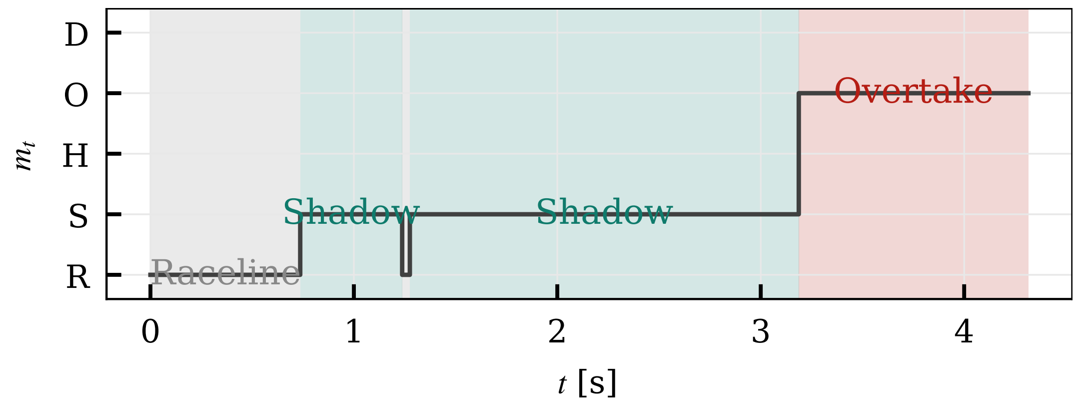


Figure 7: Case 2 FSM mode timeline.

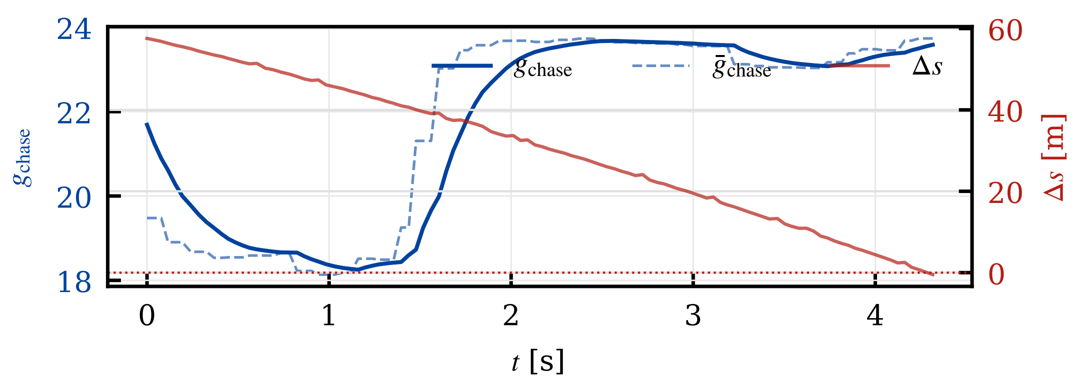


Figure 8: Case 2 chase score and relative progress.

The path figure gives the geometric view of the pass. The subsequent single-panel figures show why the maneuver is executable: the speed and acceleration profiles stay smooth, the MPCC errors remain bounded, and the FSM changes state through score-threshold crossings rather than noise.

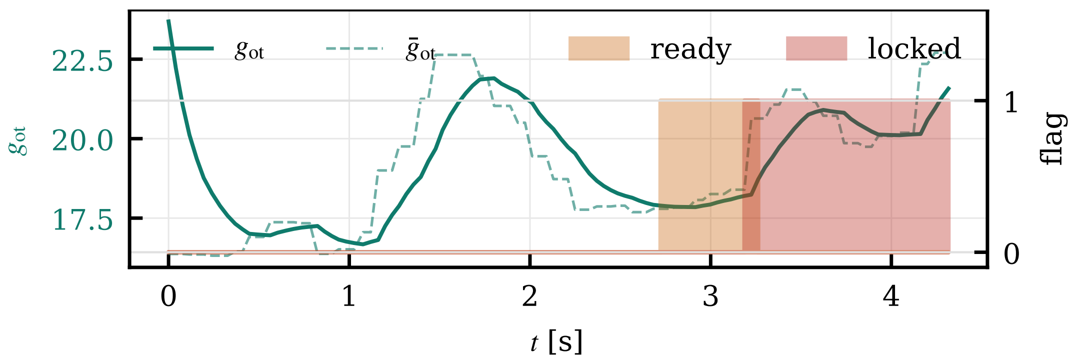


Figure 9: Case 2 overtake score and decision flags.

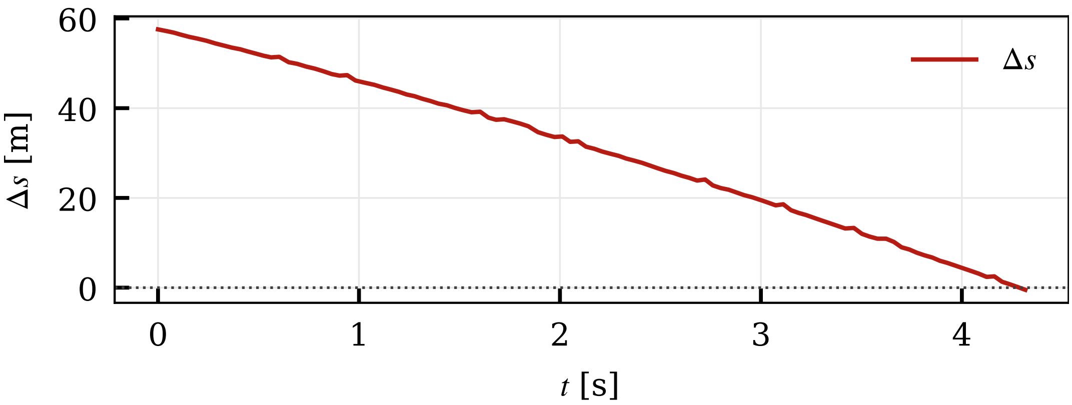


Figure 10: Case 2 relative progress.

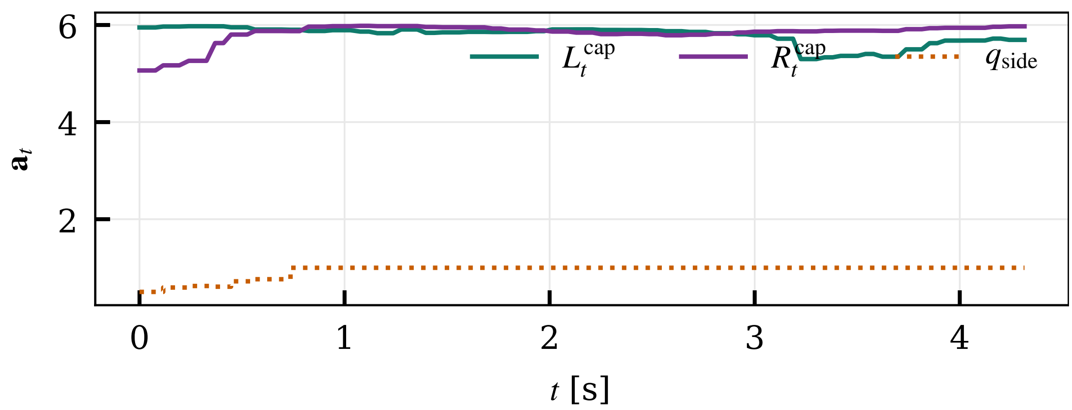


Figure 11: Case 2 corridor caps and committed side.

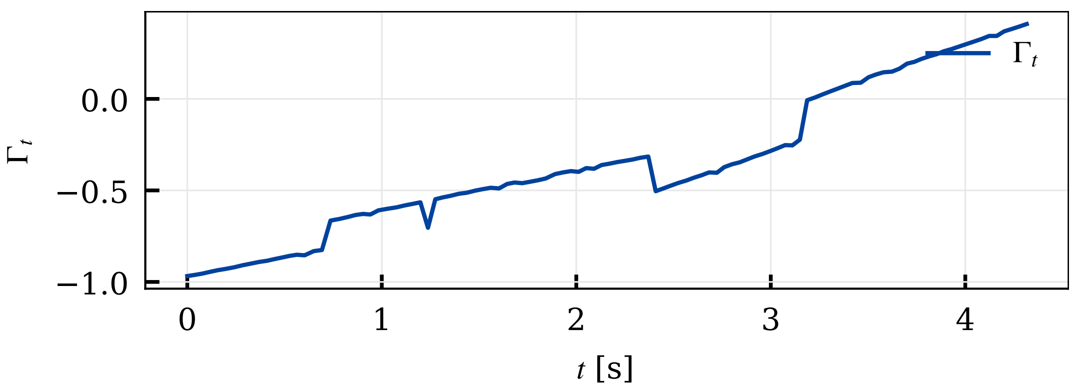
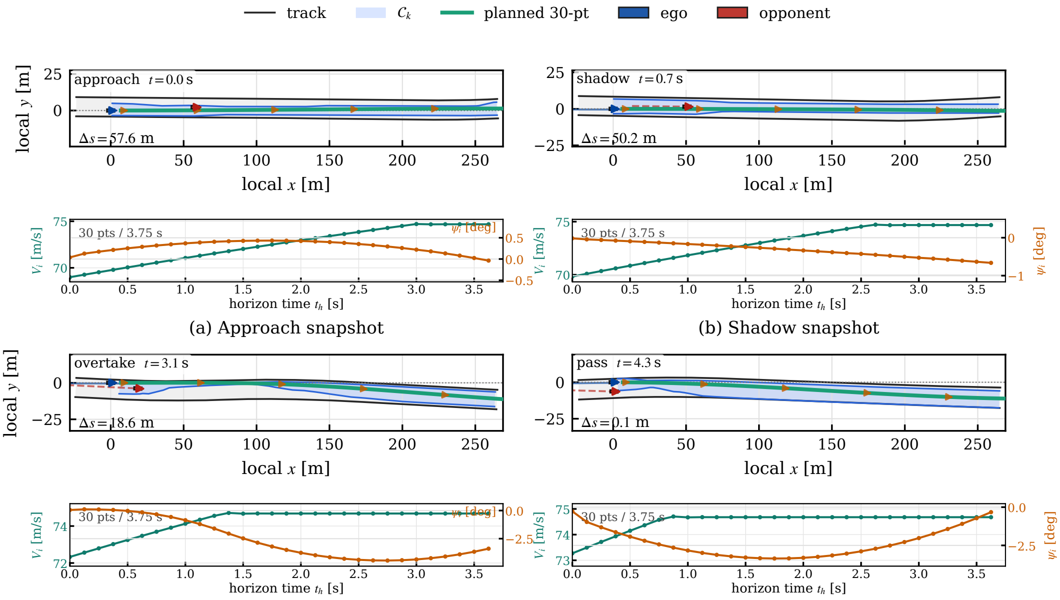


Figure 12: Case 2 tactical index.



(a) Approach snapshot (b) Shadow snapshot

(c) Overtake snapshot (d) Pass snapshot

Figure 13: Case 2 engagement snapshots. This composite figure is kept as one PDF because the four stage views are already paired with their horizon preview plots.

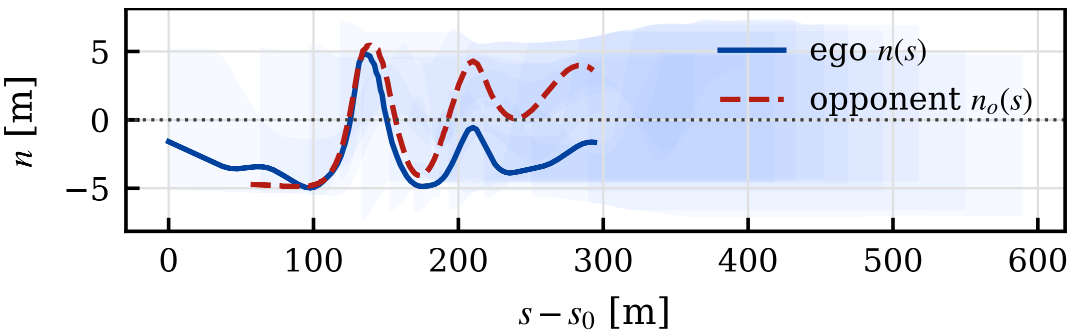


Figure 14: Case 5 Frenet path and tactical corridor. The ego trace remains inside the feasible corridor generated by the tactical layer.

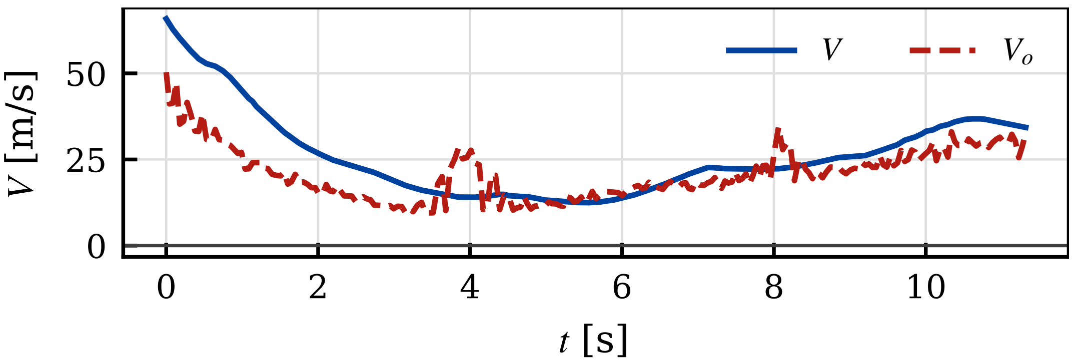


Figure 15: Case 5 speed profile.

The engagement snapshots add the spatial progression during the approach, shadow, overtake, and pass stages.

7.3 Case 5: Longer Fast-Closure Pass

This case completes the overtake with min $\Delta s = -0.84$ m, $\Delta t_{\text{pass}} = 8.63$ s, max $V = 65.83$ m/s, RMS$(e_l) = 0.05$ m, RMS$(e_v) = 1.82$ m/s, and tactical gain $\Delta\Gamma = 1.75$.

The path figure gives the geometric view of the pass. The subsequent single-panel figures show

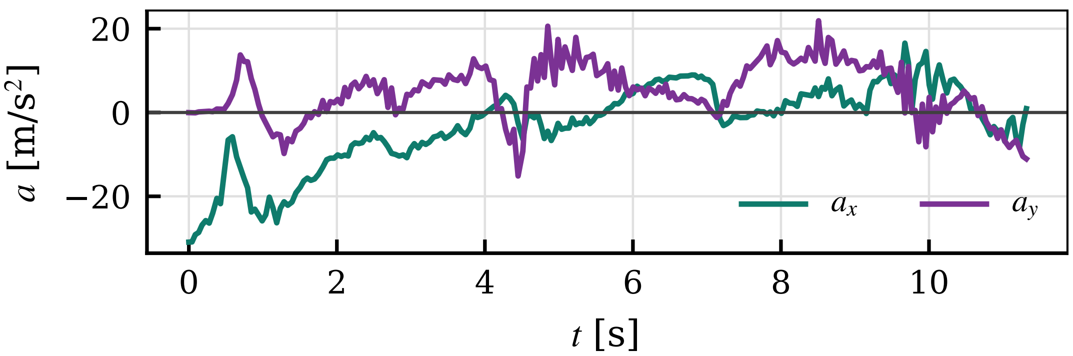


Figure 16: Case 5 longitudinal and lateral acceleration.

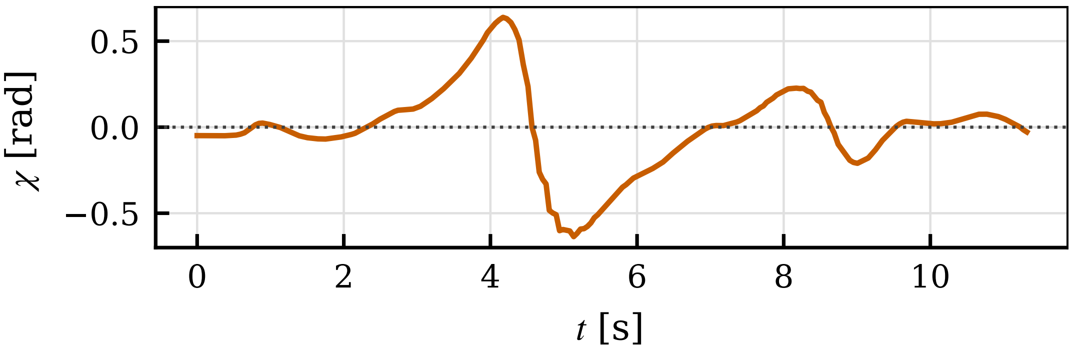


Figure 17: Case 5 heading error.

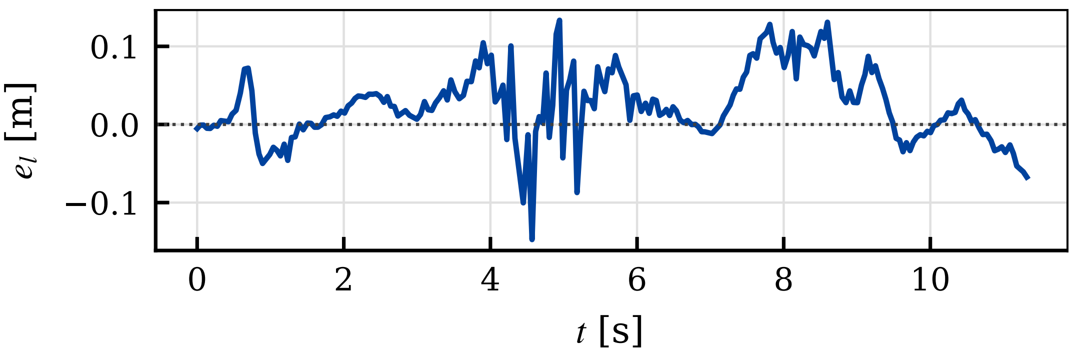


Figure 18: Case 5 lateral tracking error.

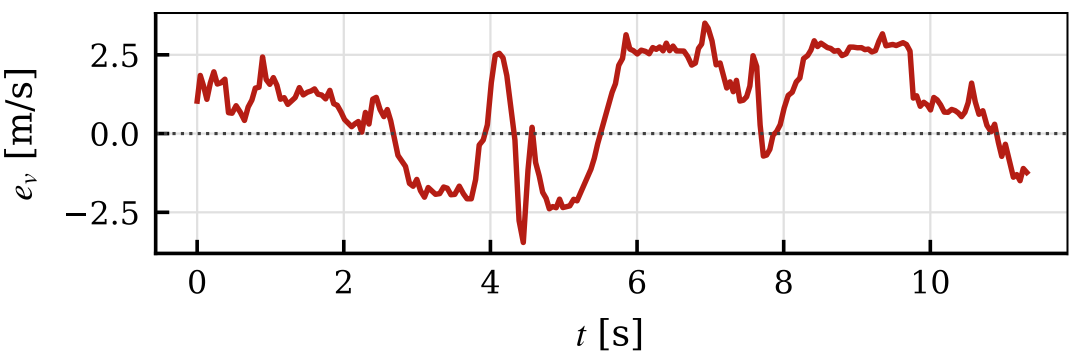


Figure 19: Case 5 speed tracking error.

why the maneuver is executable: the speed and acceleration profiles stay smooth, the MPCC errors remain bounded, and the FSM changes state through score-threshold crossings rather than noise. The engagement snapshots add the spatial progression during the approach, shadow, overtake, and pass stages.

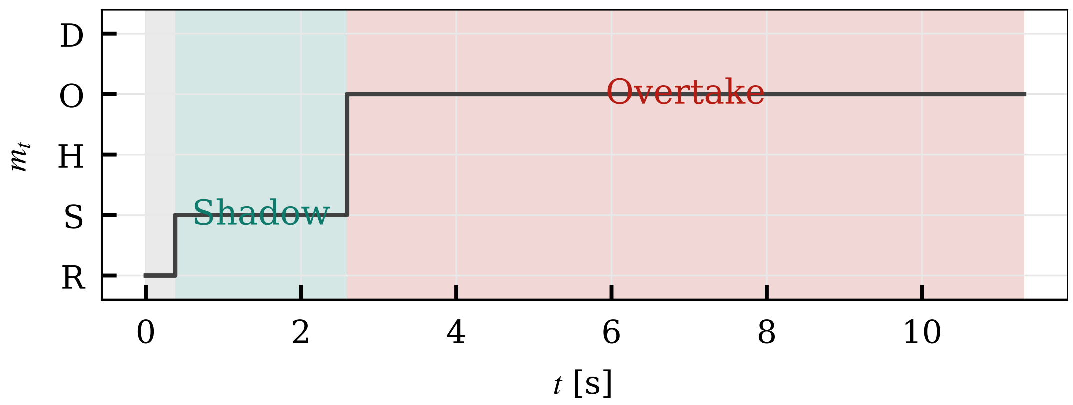


Figure 20: Case 5 FSM mode timeline.

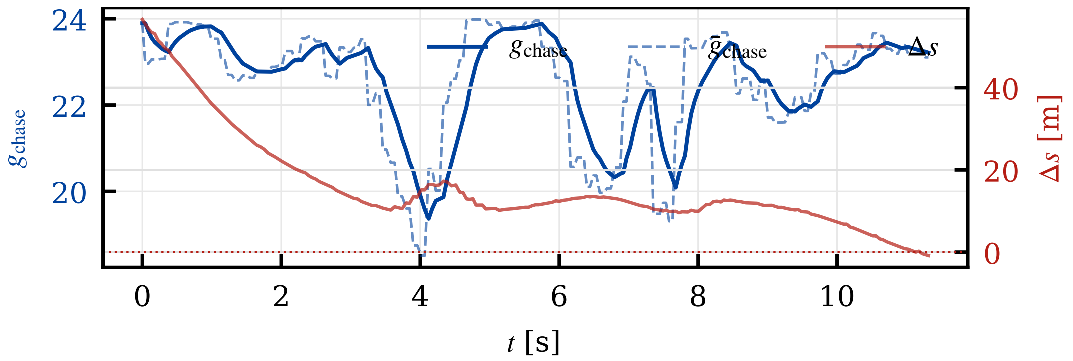


Figure 21: Case 5 chase score and relative progress.

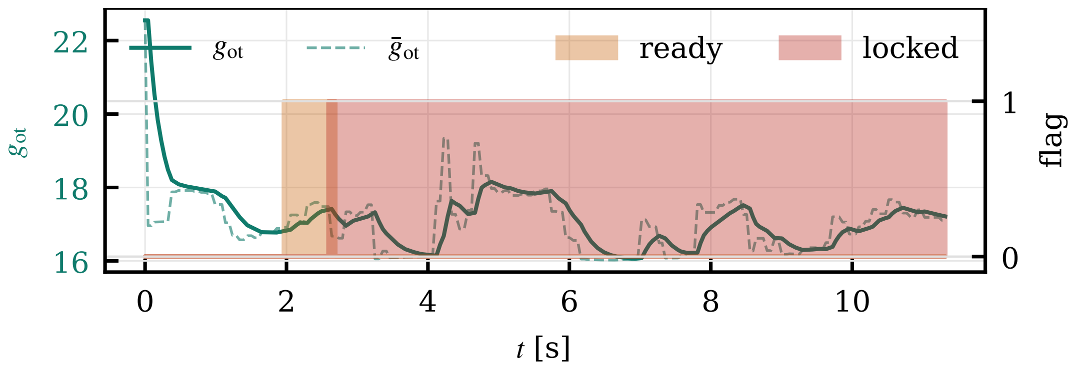


Figure 22: Case 5 overtake score and decision flags.

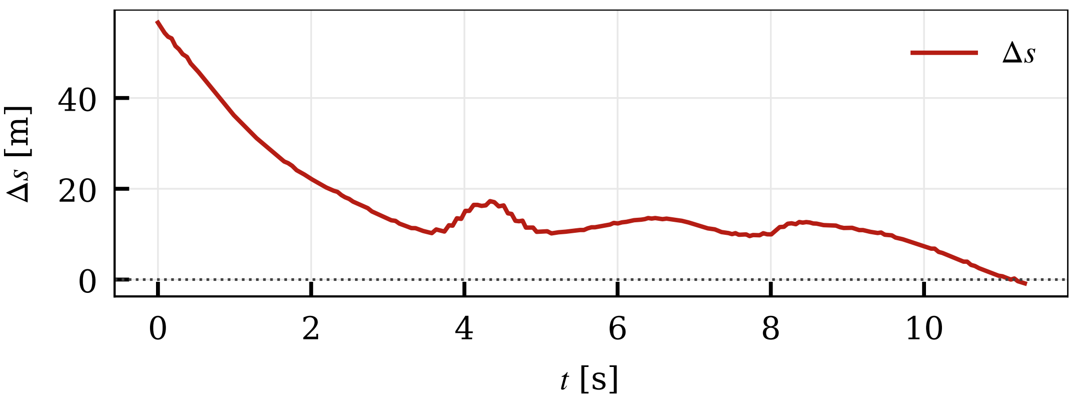


Figure 23: Case 5 relative progress.

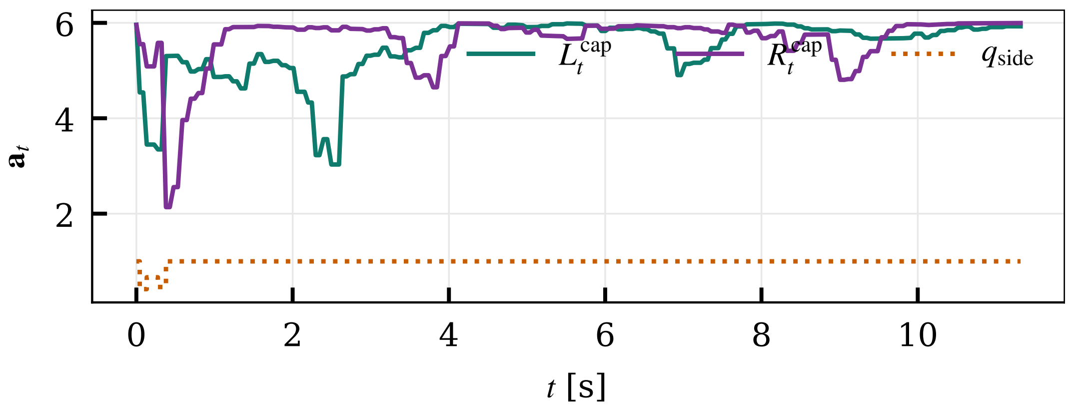


Figure 24: Case 5 corridor caps and committed side.

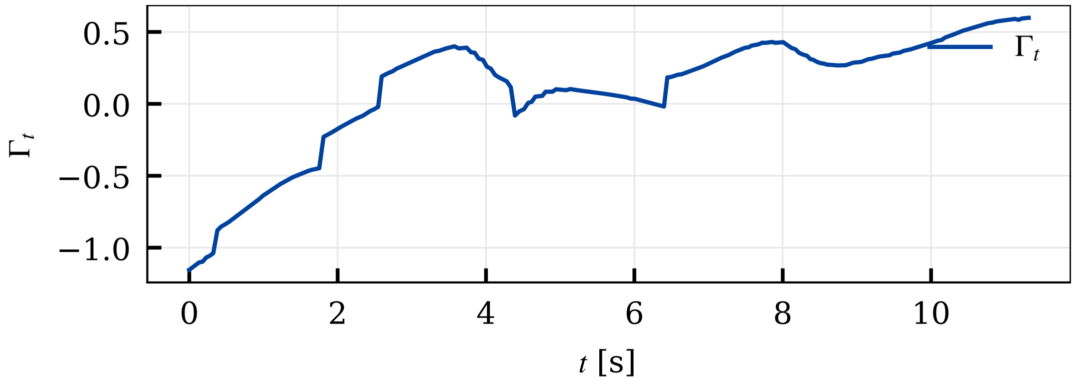


Figure 25: Case 5 tactical index.

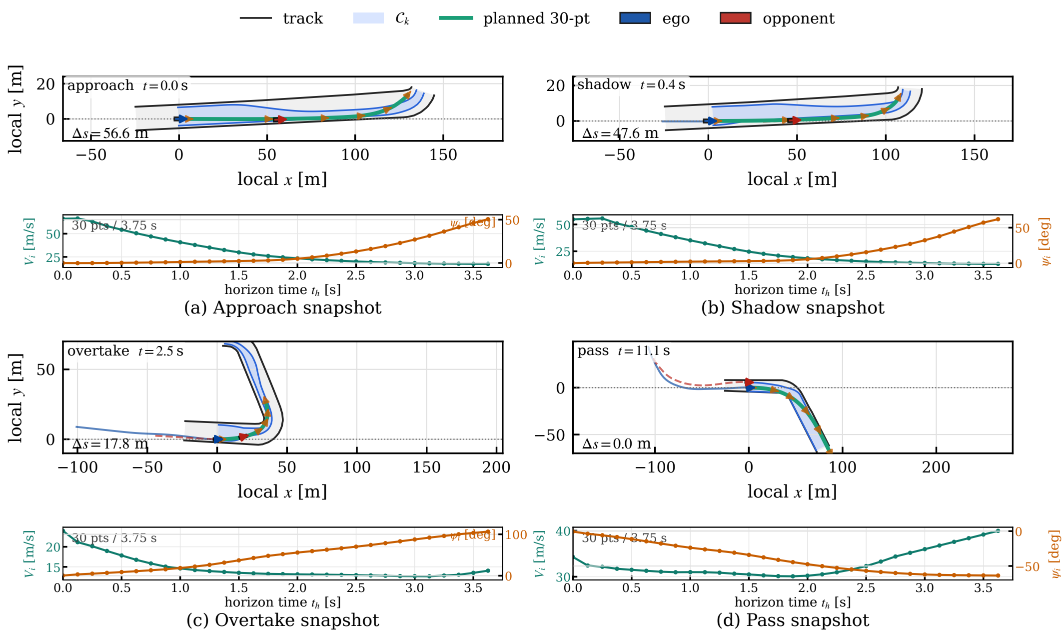


Figure 26: Case 5 engagement snapshots. This composite figure is kept as one PDF because the four stage views are already paired with their horizon preview plots.

7.4 Validation Statement

The two cases show the same pattern under different timing margins. Case 2 demonstrates that the hierarchy can make and execute a tactical commitment with very little time to react. Case 5 shows that the same commitment mechanism remains stable over a longer overtake interval with larger dynamic transients. In both runs, the learned component only changes the corridor, speed cap, and

FSM thresholds; it does not bypass the planner or controller constraints. The successful passes, stable side commitments, bounded accelerations, and sub-decimeter lateral RMS errors therefore support the main claim of the paper: tactical learning is effective when it shapes an auditable feasible set that model-based planning and control can execute.

8 Conclusion

We have described a complete tactical–planning–control stack for wheel-to-wheel autonomous racing on a Formula-class platform. A unified Frenet model derived from a dynamic single-track vehicle underpins all three layers; a TQC-driven tactical layer with an embedded track prior, a hysteretic FSM and a heading-aware corridor carver synthesizes a smooth, time-varying feasible region $\mathcal{C}_k$; a SOCP local planner with sequential linearization and a forward–backward GG-coupled velocity sweep produces curvature-smooth, grip-respecting trajectories inside $\mathcal{C}_k$; and an MPCC controller with a positive arc-length progress terminal and a soft friction-circle constraint executes them at the control rate.

The simulation study shows that this separation is more than an implementation convenience. Across two Yas Marina fast-closure overtake cases, the learned component changes only the feasible corridor and speed cap, while the unchanged planner and controller complete passes with lateral RMS tracking errors no larger than 0.05 m and peak speed up to 73.10 m/s. These results support the central design claim: tactical learning is most useful when it constrains an auditable model-based stack rather than replacing it.

Future work will expand the evaluation to multi-opponent traffic, include ablation studies of the track prior and FSM hysteresis, and transfer the validated stack from closed-loop simulation to controlled track testing. On the algorithmic side, the next extensions are opponent prediction tubes in the SOCP corridor, a conic speed-planning formulation on the GG envelope, and tactical adaptation of the MPCC progress weight q_{prog}.

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